DO NOW — RECALL

Programming Techniques · Value and Reference

Name: _____ Class: _____ Date: _____

1. What is a local variable?

2. Why use a method?

3. What does an accumulator store?

4. What is a hash collision?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A name scoped to a function or block

2. Model answer

To modularise and reuse logic

3. Model answer

A running total

4. Model answer

Two keys produce the same location

By the end, you can…

• Describe value and reference behaviour

• Trace changes through method calls

• Select suitable parameter passing

Vocabulary — know these terms

Recall the term — by value

Recall the term — ref

Recall the term — reference

Recall the term — argument

Recall the term — mutation

NEW LEARNING

Read, annotate and explain

Compare changes to a copy and an object

class Box { public int Value; } class Program { static void CopyInt(int n) { n = 9; } static void Mutate(Box p) { p.Value = 9; } static void Replace(ref Box p) { p = new Box { Value = 20 }; } }

Explain the important decision

An int parameter normally receives a copied number. A Box parameter normally receives a copied reference to the same object. ref instead aliases the caller variable, allowing that variable to be reassigned.

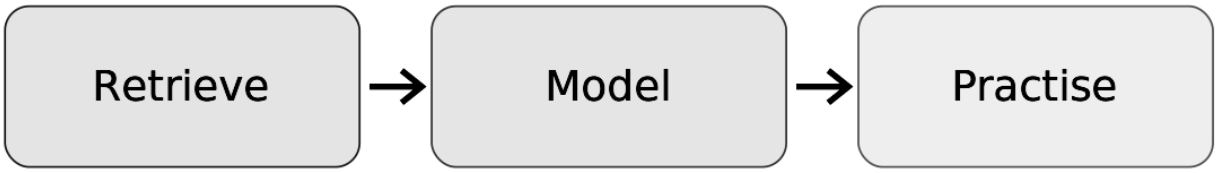
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Compare changes to a copy and an object

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Three-stage diagram showing retrieve, model, and practise

I DO — WORKED EXAMPLE

Follow the worked method

Compare changes to a copy and an object

Copy an integer value

Start x=5; CopyInt(x) changes only local n to 9. The caller x remains 5. This is pass-by-value.

Copy a reference; mutate one object

Start item→Box A(Value=5), alias=item. Mutate(item) copies the reference, then changes Box A.Value to 9. Both item and alias still refer to Box A.

Reassign the caller through ref

Replace(ref item) makes caller item refer to new Box B(Value=20). alias still refers to Box A(Value=9). Without ref, assigning a new Box to local p would leave item unchanged.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

If a method takes ref int n and adds 2, starting 3, what is the caller value?

Method / working:

2. Problem 2

Does passing a reference-type object by value copy the object itself?

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Problem 1 If a method takes ref int n and adds 2, starting 3, what is the caller value? Discuss, then respond

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Problem 2 Does passing a reference-type object by value copy the object itself?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Explain why changing a normal int parameter does not change the original. [2]

Question 2

Apply independently
Write a C# method header to increment an integer by reference. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Distinguish passing an object reference by value from using ref. [3]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: a reference-type argument passed normally allows the method to replace the caller's variable.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check your thinking

5.

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Discuss, then respond

Does passing a reference-type object by value copy the object itself?

Check your thinking

No; the reference is copied.

IMPROVE: Explain one correction to your own reasoning.